

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

D. Madhu Sudhan
192524214

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07 20

COURSE OUTCOME COVERED

CO2: Configure IP-addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%
Software: Cisco Packet Tracer, Wireshark

QUESTIONS: 5

Total Marks: 50

Annexure A

Debugging & Optimisation Task

81.7.4

Code of Conduct:

I, D. Madhu Sudhan ¹⁹²⁵²⁴²¹⁴ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

D. Madhu Sudhan

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:3329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network 192.168.1.0/24 into several departments using the subnet masks /26, /27, and /28. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
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- Debugging the addressing plan by locating subnet allocation mistakes.

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5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

	Problem Identification (20%)	Debugging (20%)	Optimization (25%)	Analysis (20%)	Code Quality (15%)
Q1	1.5	2	2	1	1 7.5
Q2	1.5	2	1.5	1.5	1 7.5
Q3	2	2.5	1.5	1.5	1 8.5
Q4	2	2	2	2	1.5 9.5
Q5	1.5	1.5	1	2	1.5 7.5

40.5

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding(2)	Identifies most errors with minor gaps(1.5)	Identifies some errors but misses key issues(1)	Unable to identify errors correctly(0)
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues(2.5)	Uses appropriate debugging methods with minor errors(2)	Limited debugging approach; requires guidance(1.5)	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms(2)	Improves time complexity with minor gaps in efficiency(1.5)	Limited improvement in time complexity(1)	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions(2)	Shows reasonable analysis with some justification(1.5)	Limited analysis; weak reasoning(1)	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation(1.5)	Code is mostly clear with minor documentation gaps(1)	Code lacks structure and proper documentation(0.5)	Poorly written code with no documentation

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Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
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2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
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Although both devices are connected to the same network, communication fails.

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3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

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- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

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Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
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RUBRICS FOR CALCULATION

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- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.

Calculating the total number of usable host addresses after correction

IPv4 Subnetting – Debugging and Optimisation

Given:

- **IP Address: 192.168.10.65**
- **Subnet Mask: 255.255.255.192 (/26)**
- **Default Gateway: 192.168.10.1**

Answer:

The IP address 192.168.10.65 belongs to the subnet 192.168.10.64/26.

Parameter	Value
Network Address	192.168.10.64
Broadcast Address	192.168.10.127
Valid Host Range	192.168.10.65 – 192.168.10.126
Configured IP	192.168.10.65 – Valid
Configured Gateway	192.168.10.1 – Invalid

The main configuration error is that 192.168.10.65 and 192.168.10.1 belong to different /26 subnets. Therefore, the default gateway is incorrect for the configured IP address.

Optimized configuration:

- **IP Address: 192.168.10.66**
- **Subnet Mask: 255.255.255.192**
- **Default Gateway: 192.168.10.65**

For a /26 subnet:

Total addresses = 64

Usable host addresses = 62

Therefore, the corrected subnet provides 62 usable host addresses.

2. Two devices are configured with the following IPv6 addresses:

- **Device A: 2001:db8::ff00:42:8329/64**
- **Device B: 2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

IPv6 Addressing – Debugging and Optimisation

Given:

- **Device A: 2001:db8::ff00:42:8329/64**
- **Device B: 2001:db8:0:0:1::1/64**

Expanded addresses:

- **Device A: 2001:0db8:0000:0000:0000:ff00:0042:8329/64**
- **Device B: 2001:0db8:0000:0000:0001:0000:0000:0001/64**

Network prefixes:

- **Device A: 2001:db8:0:0::/64**
- **Device B: 2001:db8:0:0:1::/64**

The devices do not belong to the same subnet. The problem is an IPv6 prefix mismatch.

Optimized configuration:

- **Device A: 2001:db8:0:0::1/64**
- **Device B: 2001:db8:0:0::2/64**

Both devices will then belong to 2001:db8:0:0::/64.

A /64 IPv6 subnet contains:

$2^{64} = 18,446,744,073,709,551,616$ addresses

Therefore, the corrected subnet has 2^{64} possible interface addresses.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

IPv4 Subnet Allocation and VLSM Optimisation

The allocated network is 192.168.1.0/24.

/26 Subnet

- Network Address: 192.168.1.0
- Broadcast Address: 192.168.1.63
- Host Range: 192.168.1.1 – 192.168.1.62
- Total Addresses: 64
- Usable Addresses: 62

/27 Subnet

- Network Address: 192.168.1.64
- Broadcast Address: 192.168.1.95
- Host Range: 192.168.1.65 – 192.168.1.94
- Total Addresses: 32
- Usable Addresses: 30

/28 Subnet

- Network Address: 192.168.1.96
- Broadcast Address: 192.168.1.111
- Host Range: 192.168.1.97 – 192.168.1.110
- Total Addresses: 16
- Usable Addresses: 14

Subnet	Network	Broadcast	Usable Hosts
/26	192.168.1.0	192.168.1.63	62
/27	192.168.1.64	192.168.1.95	30
/28	192.168.1.96	192.168.1.111	14

A /26 can be inefficient for a small department because it provides 62 usable addresses. A /28 may be insufficient for a department requiring more than 14 hosts.

Common addressing mistakes include overlapping subnet ranges, assigning network/broadcast addresses to hosts, and assigning a subnet that is too small for the department.

Since the question does not specify the exact number of hosts required by each department, an exact VLSM allocation cannot be uniquely determined.

An example optimized VLSM allocation is:

Requirement	Subnet	Network	Broadcast	Usable Hosts
Up to 126 hosts	/25	192.168.1.0	192.168.1.127	126
Up to 62 hosts	/26	192.168.1.128	192.168.1.191	62
Up to 30 hosts	/27	192.168.1.192	192.168.1.223	30
Up to 14 hosts	/28	192.168.1.224	192.168.1.239	14

VLSM improves address utilization by assigning larger subnets to departments with greater requirements and smaller subnets to departments with fewer requirements.

4. . Two devices are configured with the following IPv6 addresses:

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Although both devices are connected to the same network, communication fails.

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- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

IPv6 Addressing – Debugging and Optimisation

Device A:

2001:0db8:0000:0000:0000:ff00:0042:8329/64

Device B:

2001:0db8:0000:0000:0001:0000:0000:0001/64

Network Prefixes:

- Device A: **2001:db8:0:0::/64**
- Device B: **2001:db8:0:0:1::/64**

The two devices are not in the same subnet because their /64 network prefixes are different.

Corrected configuration:

- Device A: **2001:db8:0:0::1/64**
- Device B: **2001:db8:0:0::2/64**

Corrected subnet:

2001:db8:0:0::/64

Available IPv6 addresses:

$2^{64} = 18,446,744,073,709,551,616$

Therefore, the corrected /64 subnet provides 2^{64} possible interface addresses.

5. A router contains the following routing table.

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192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

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- Identifying the route that will be used if no matching destination exists.

Routing Table – Longest Prefix Match

Given Routing Table:

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Destination IP: 192.168.2.100

The destination 192.168.2.100 belongs to the network:

192.168.2.0/24

The matching routing entry is:

192.168.2.0/24 → 10.0.0.3

Therefore, the selected next-hop router is 10.0.0.3.

The destination IP is a valid host address in the 192.168.2.0/24 subnet.

Based on the given routing table, there is no obvious routing-table error for this destination. If communication still fails, the reachability of 10.0.0.3 and the routing beyond that router should be checked.

The specific 192.168.2.0/24 route should be retained because it provides a more specific route than the default route.

If there is no matching specific route, the router uses:

Default Route → 10.0.0.1